ЗЕРТХАНАЛЫҚ ЖҰМЫСТЫ орындауға арналған
ӘДІСТЕМЕЛІК НҰСҚАУЛЫҚ

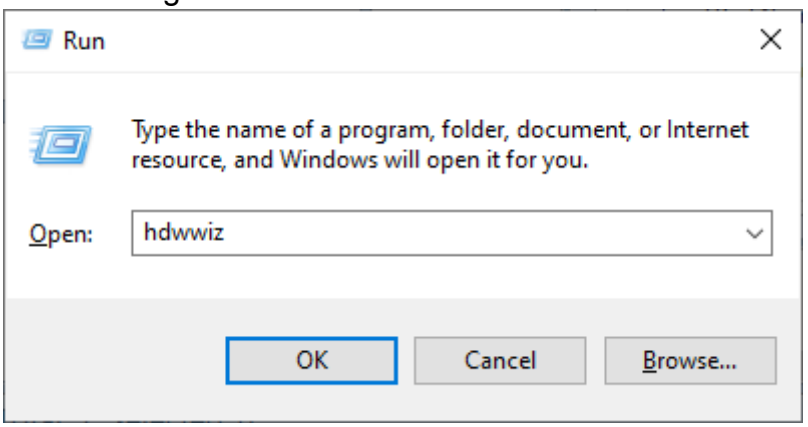
Huawei eNSP симуляторын қолданған жағдайда

Тақырыбы: Network Automation. Python
Жұмыстың мақсаты:
Жұмыстың орындалу қадамы:

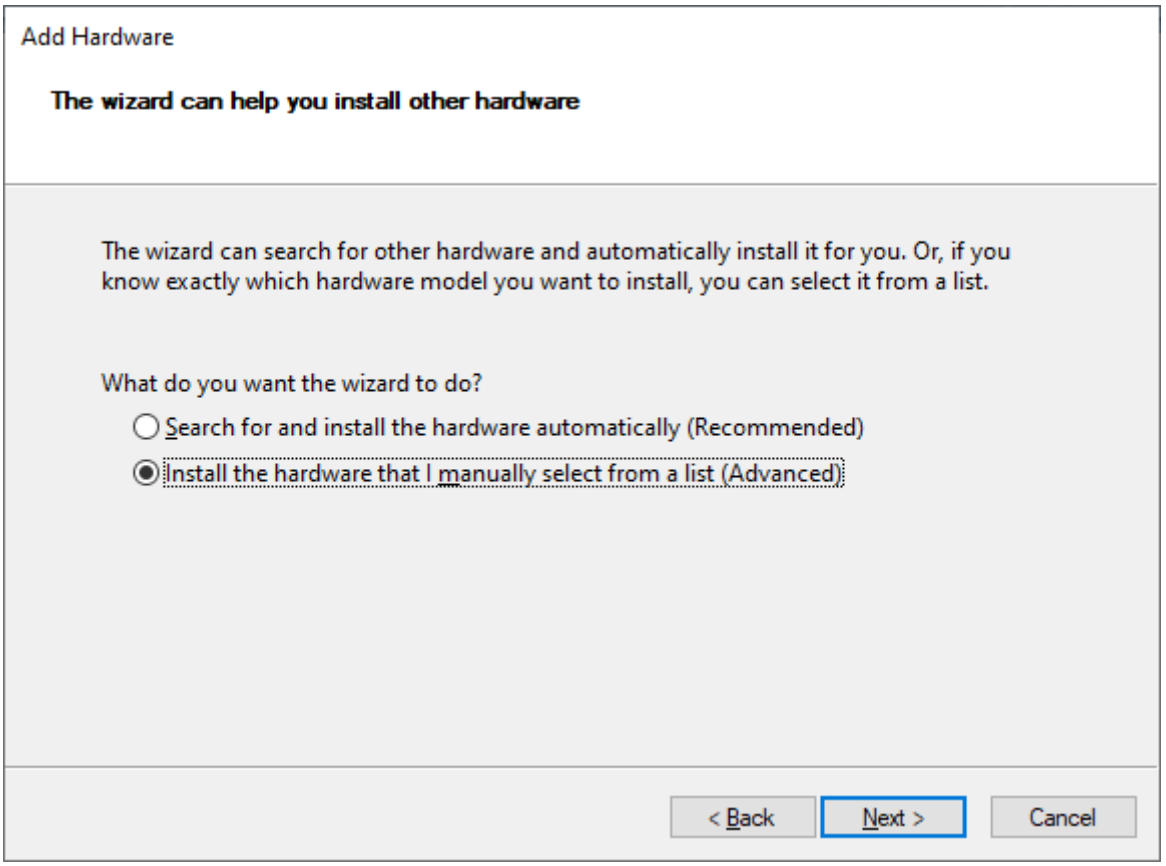
1) Windows операциялық жүйесінде Loopback interface құру
2) Windows операциялық жүйесімен eNSP симуляторын байланыстыру
3) Ubuntu Linux дистрибутивімен eNSP симуляторын байланыстыру
4) Python TELNET Library
5) Python Netmiko (Paramiko) Library
6) Python NAPALM Library

1-қадам: Windows операциялық жүйесінде Loopback interface құру

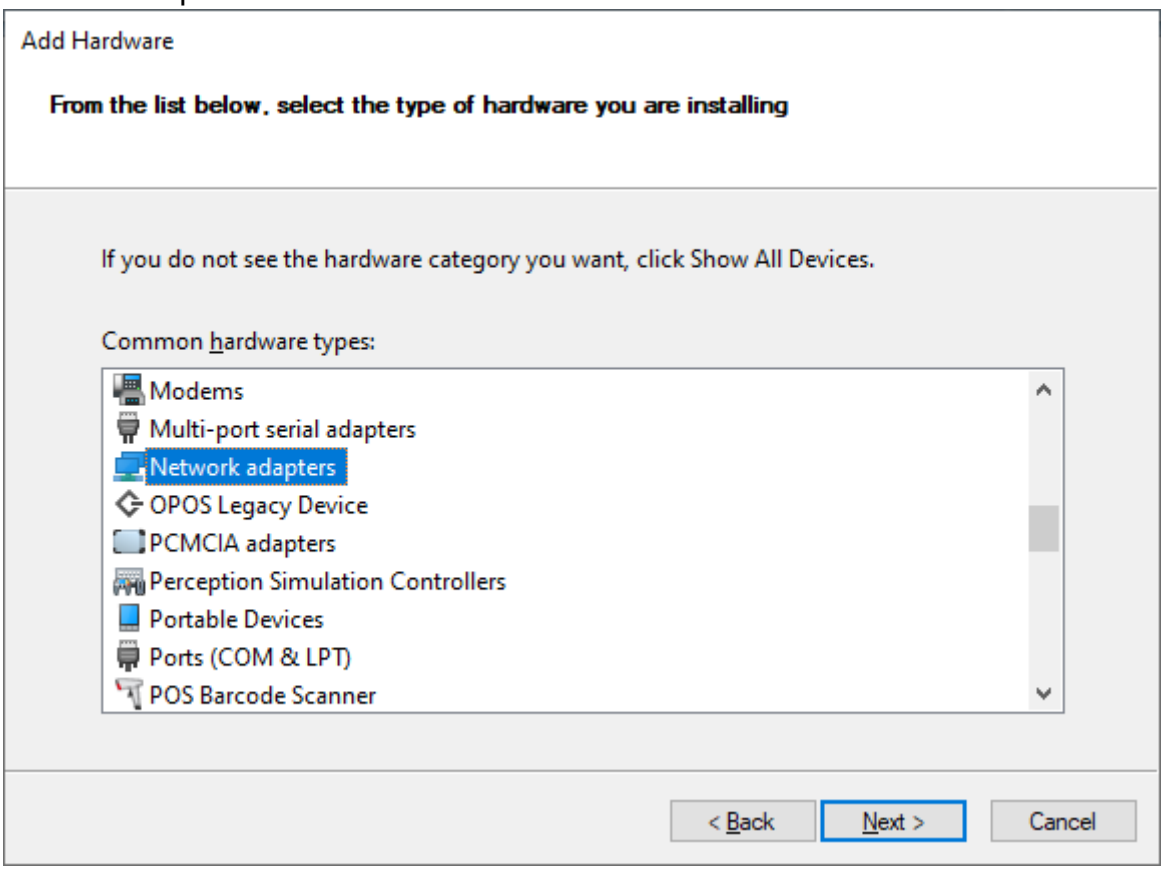
Windows logo +R



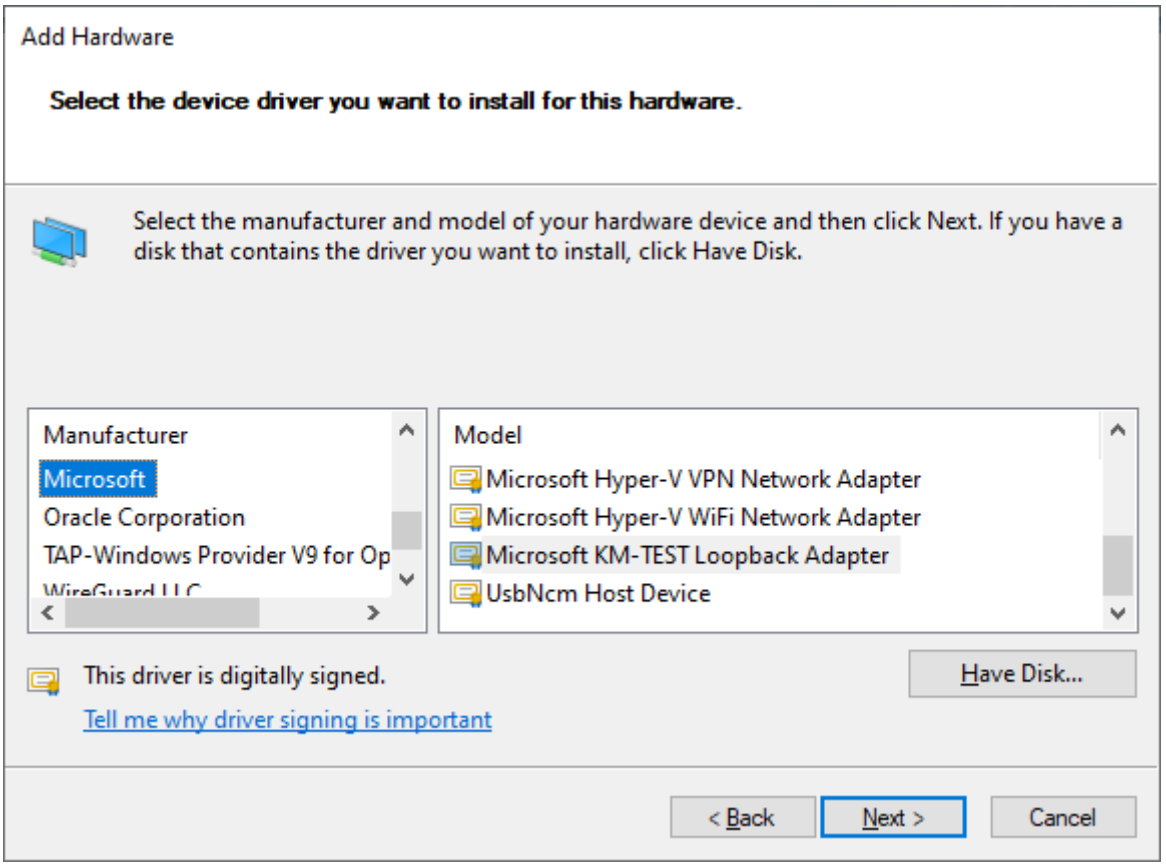
install the hardware ...



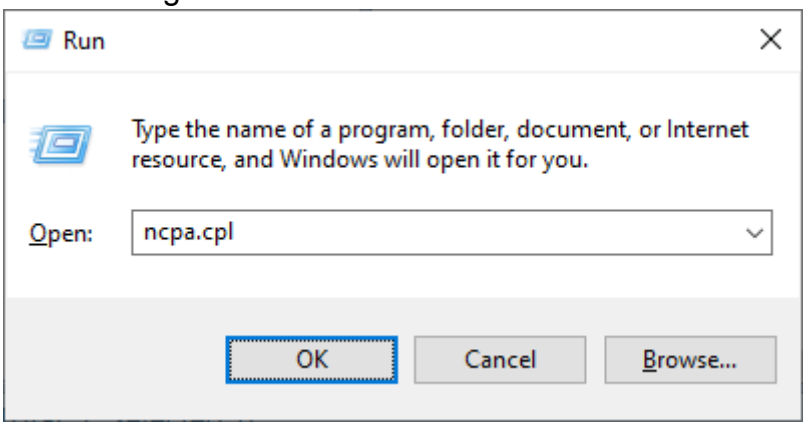
Network adapters



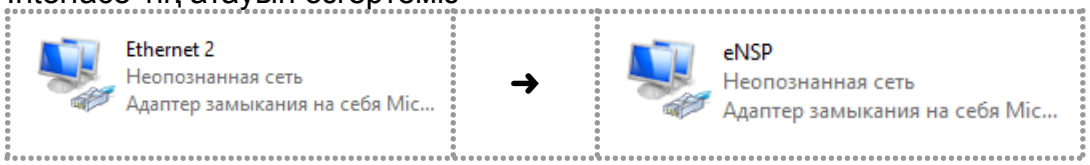
Microsoft ➜ Microsoft KM-TEST Loopback Adapter



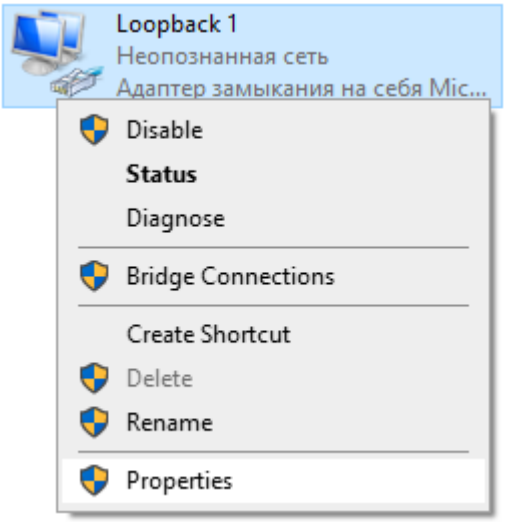
Windows logo +R

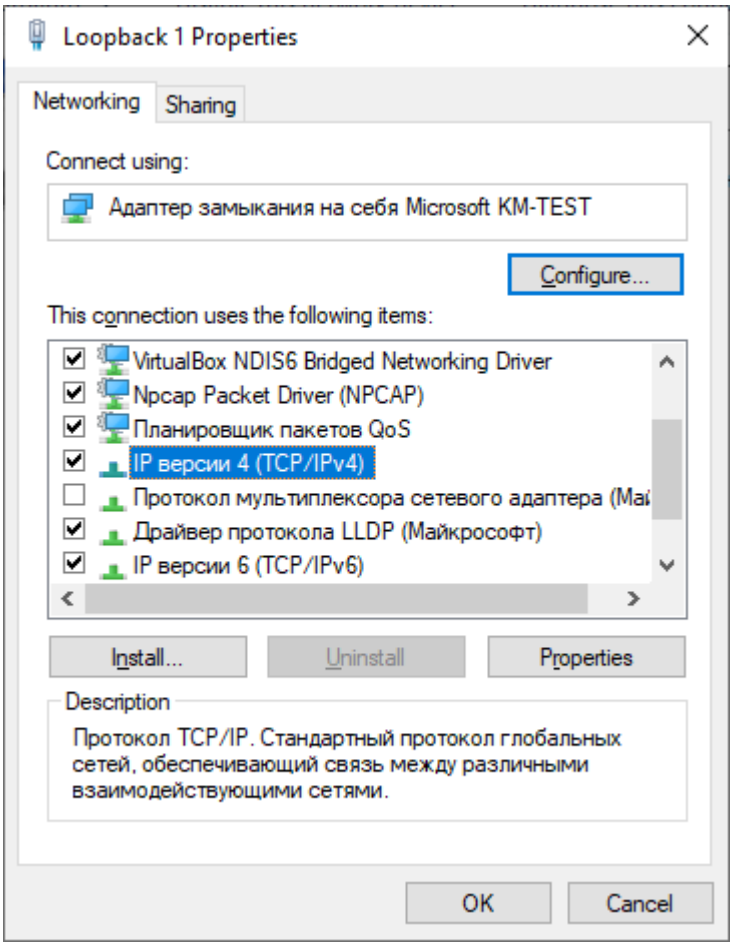


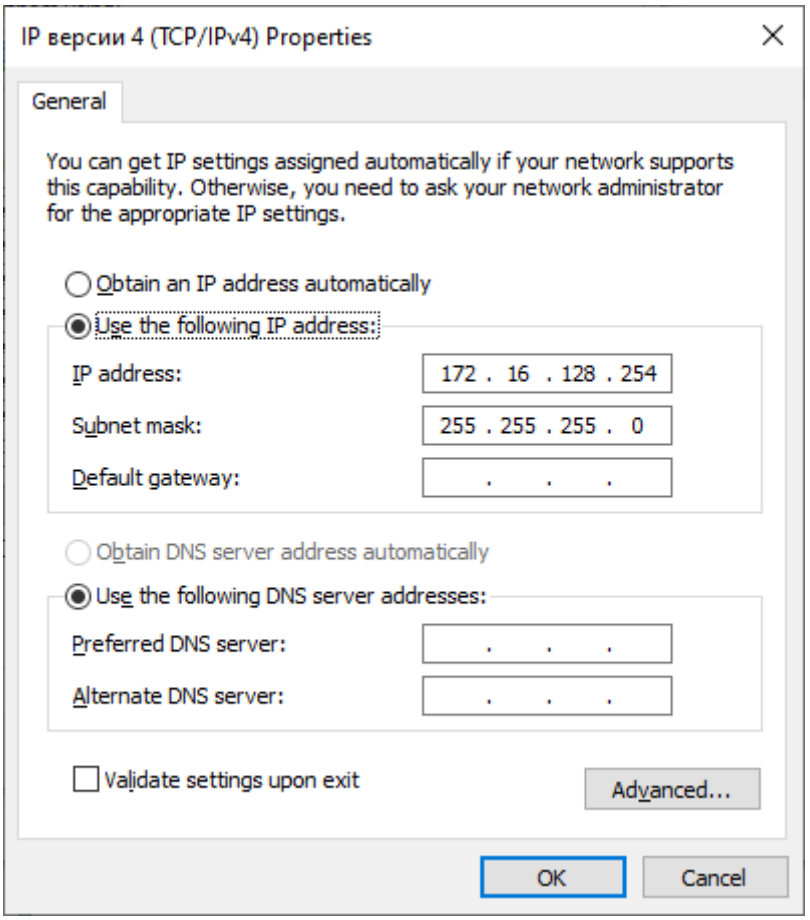
Interface-тің атауын өзгертеміз



Loopback interface ➜ Properties ➜ TCP/IPv4 ➜ IP address енгізу

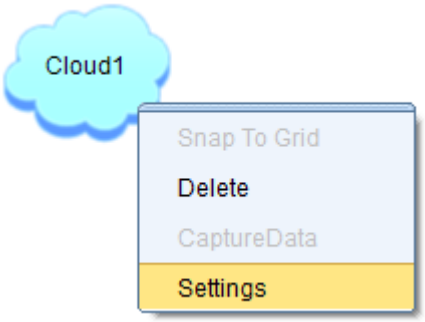

Loopback 1
Неопознанная сеть
Адаптер замыкания на себя Mic...
Disable
Status
Diagnose
Bridge Connections
Create Shortcut
Delete
Rename
Properties


Loopback 1 Properties
Networking
Sharing
Connect using:
Адаптер замыкания на себя Microsoft KM-TEST
Configure...
This connection uses the following items:
VirtualBox NDIS6 Bridged Networking Driver
Npcap Packet Driver (NPCAP)
Планировщик пакетов QoS
IP версии 4 (TCP/IPv4)
Протокол мультиплексора сетевого адаптера (Mai
Драйвер протокола LLDP (Майкрософт)
IP версии 6 (TCP/IPv6)
Install...
Uninstall
Properties
Description
Протокол TCP/IP. Стандартный протокол глобальных сетей, обеспечивающий связь между различными взаимодействующими сетями.
OK
Cancel

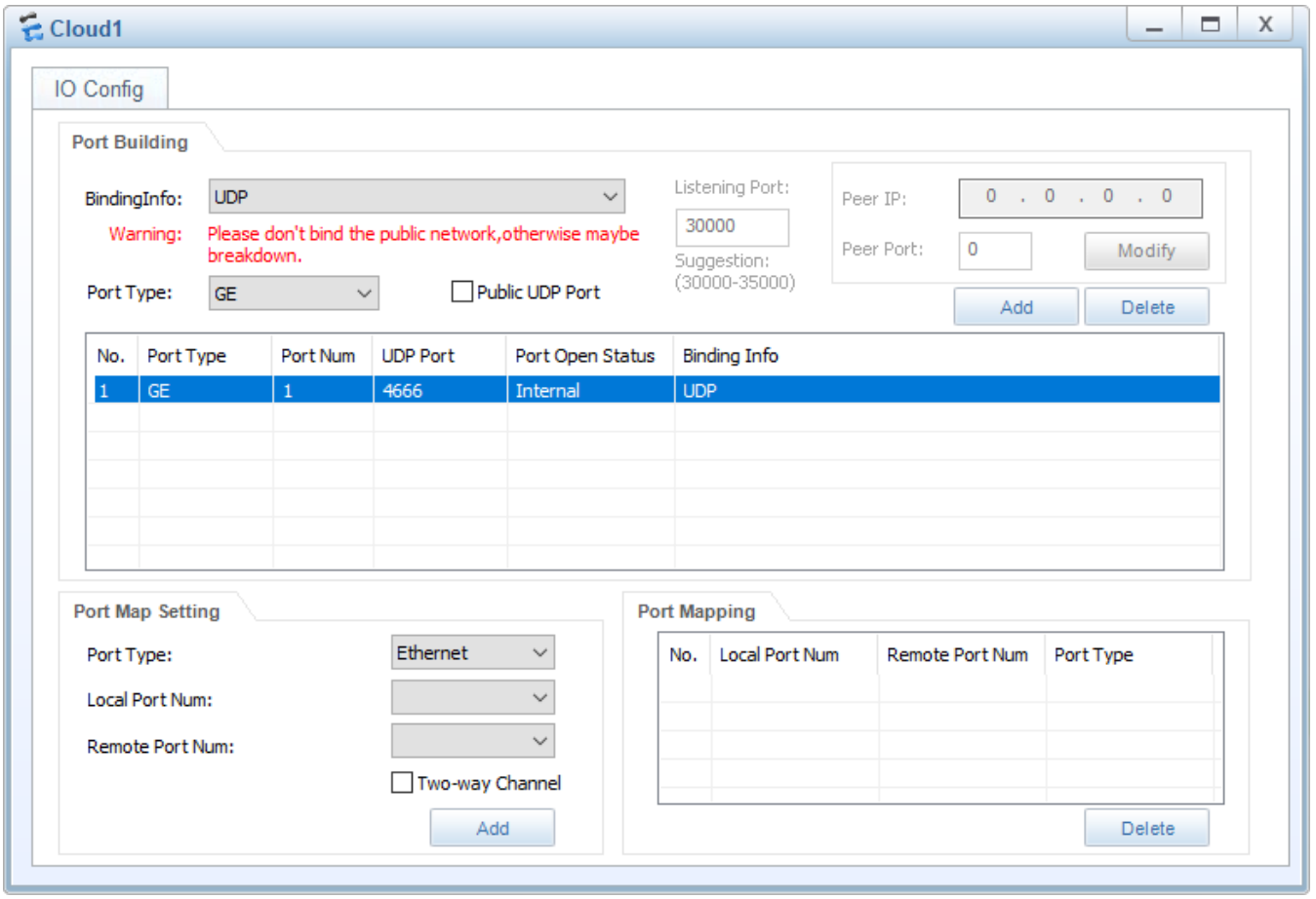


PC/Laptop ➜ Reboot

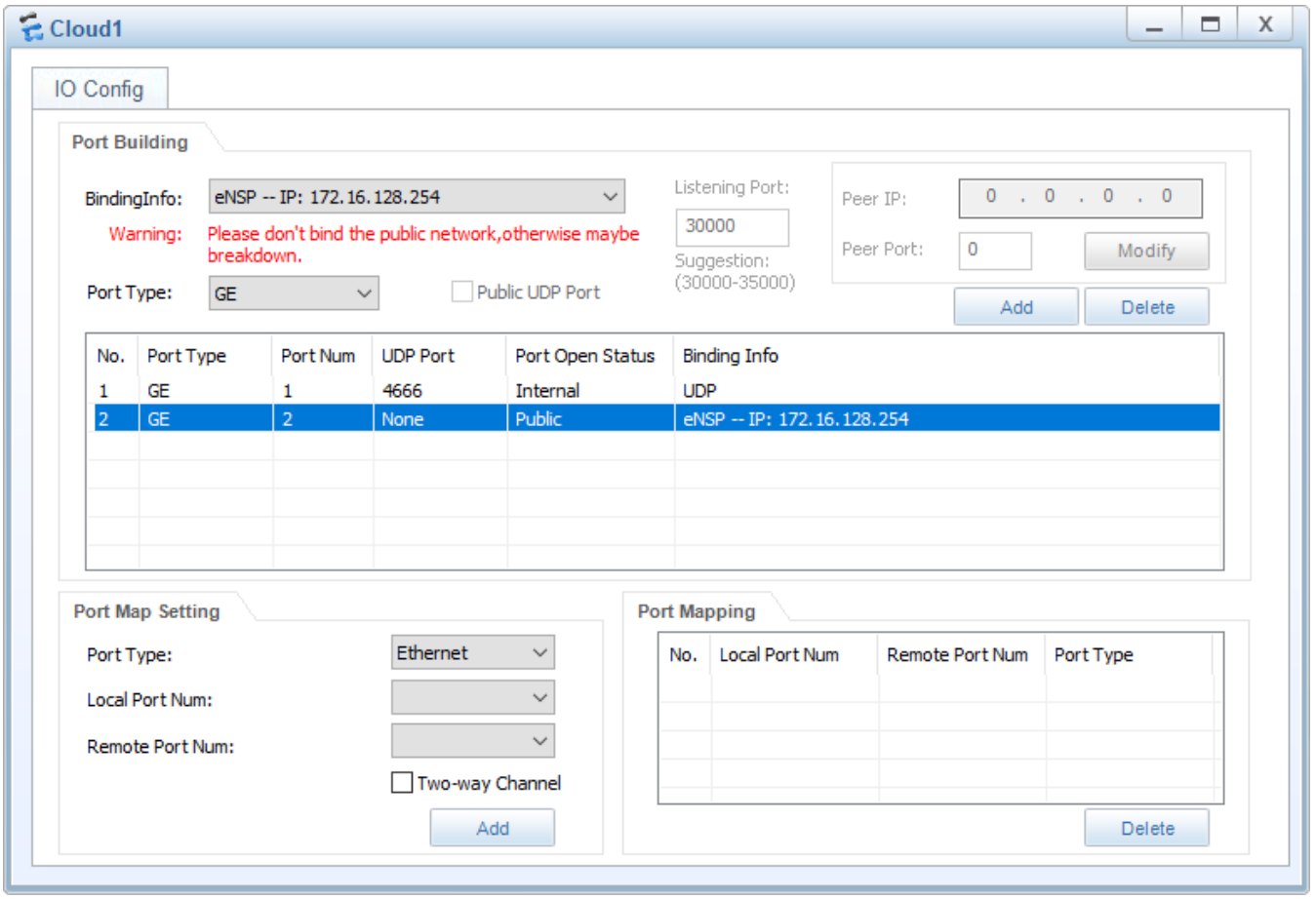
2-қадам: Windows операциялық жүйесімен eNSP симуляторын байланыстыру



Port Building ➜ BindingInfo: UDP
➜ Port Type: GE
➜ “Add” батырманы басу



Port Building:
➜ BindingInfo: eNSP
➜ Port Type: GE
➜ “Add” батырманы басу



Port Map Setting:

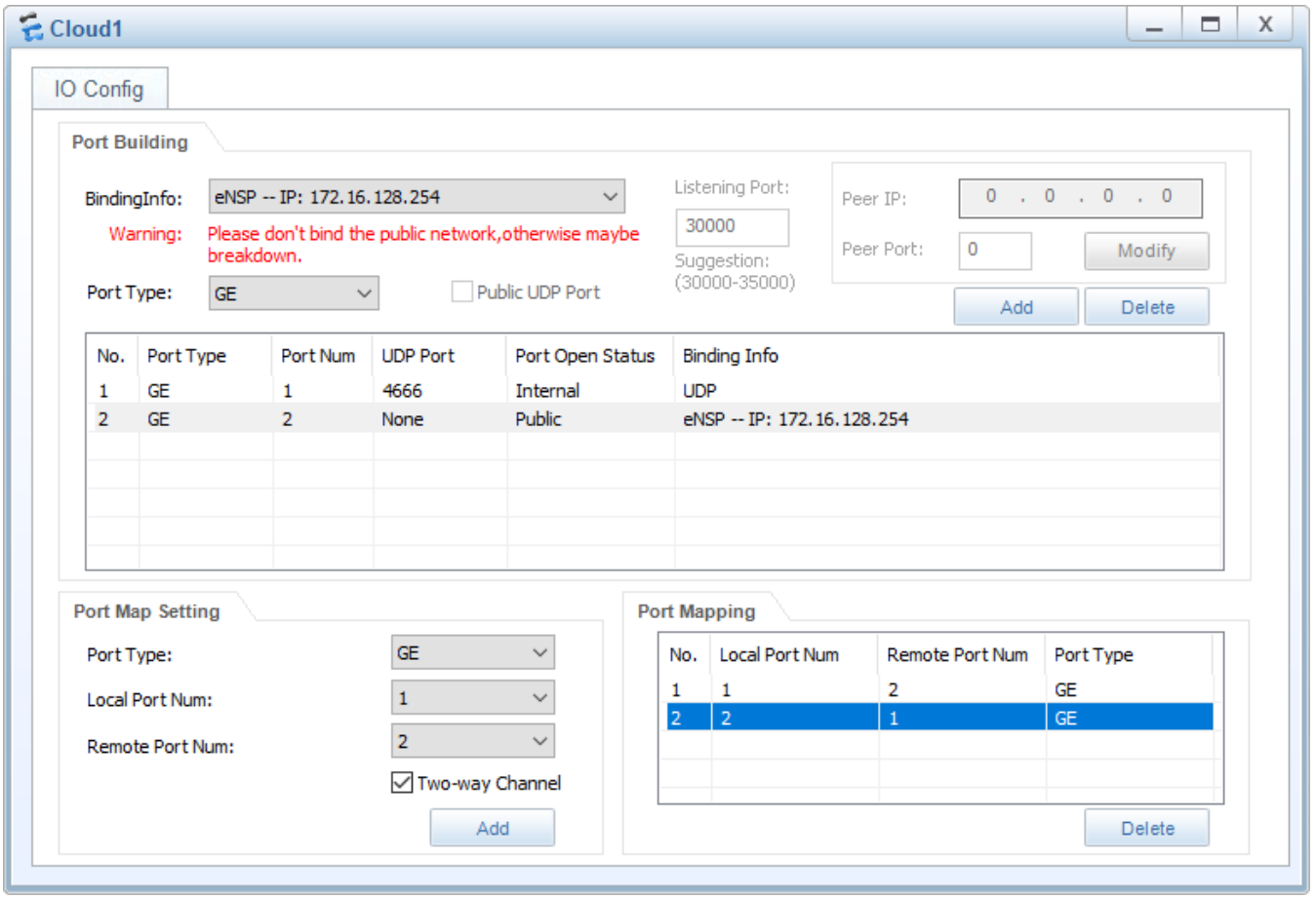
➜ Port Type: GE

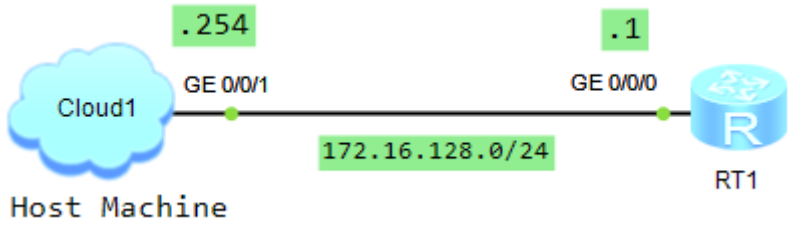
➜ Local Port Num: 1

➜ Remote Port Num: 2

☑ Two-way Channel

➜ “Add” батырманы басу



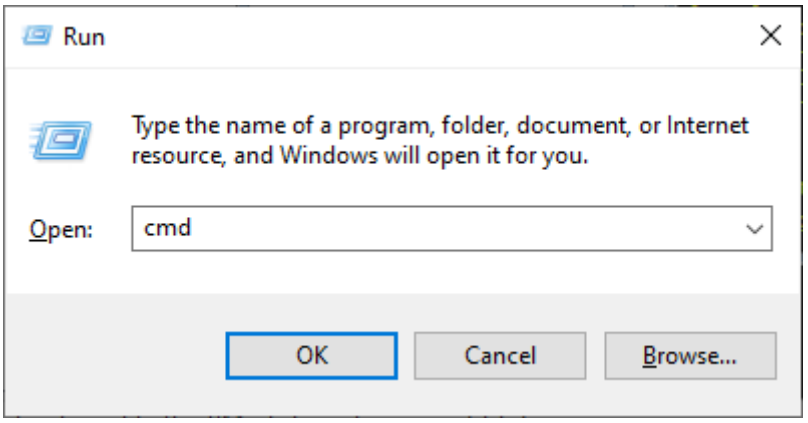


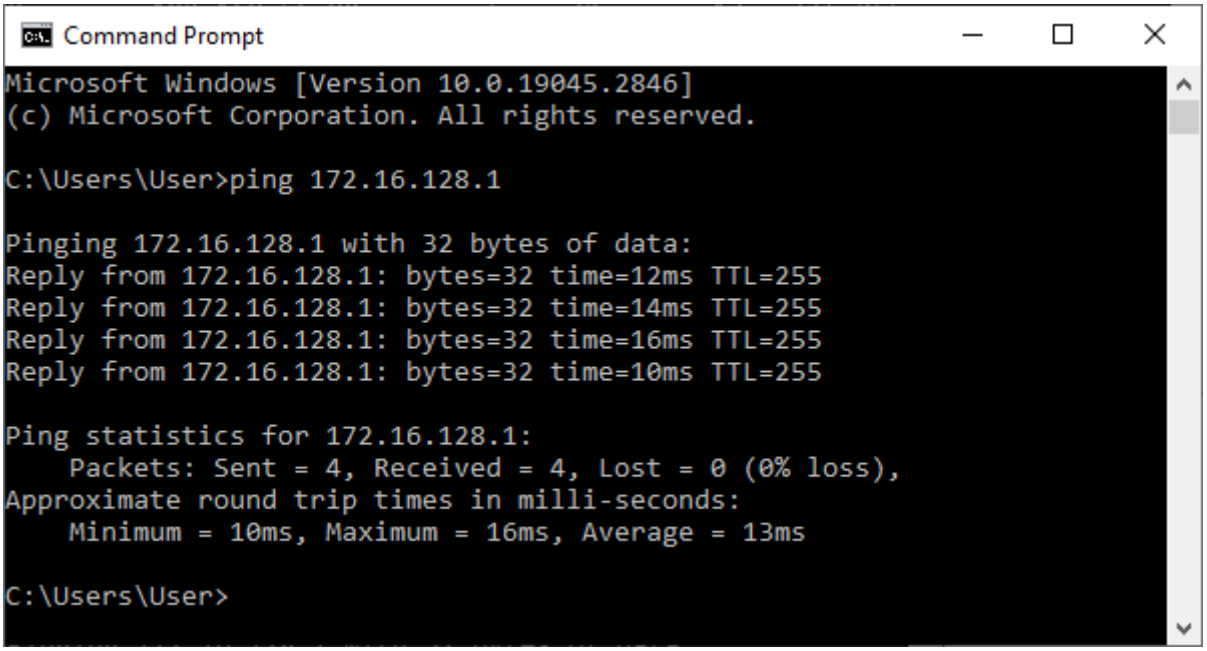

```
<Huawei> undo terminal logging
<Huawei> system-view
[Huawei] sysname RT1

[RT1] interface g0/0/0
[RT1-GigabitEthernet0/0/0] ip address 172.16.128.1 24

[RT1] display ip int brief
Interface                         IP Address/Mask      Physical   Protocol
GigabitEthernet0/0/0              172.16.128.1/24      up         up
GigabitEthernet0/0/1              unassigned           down       down
GigabitEthernet0/0/2              unassigned           down       down
NULL0                             unassigned           up         up(s)
```

Windows logo +R



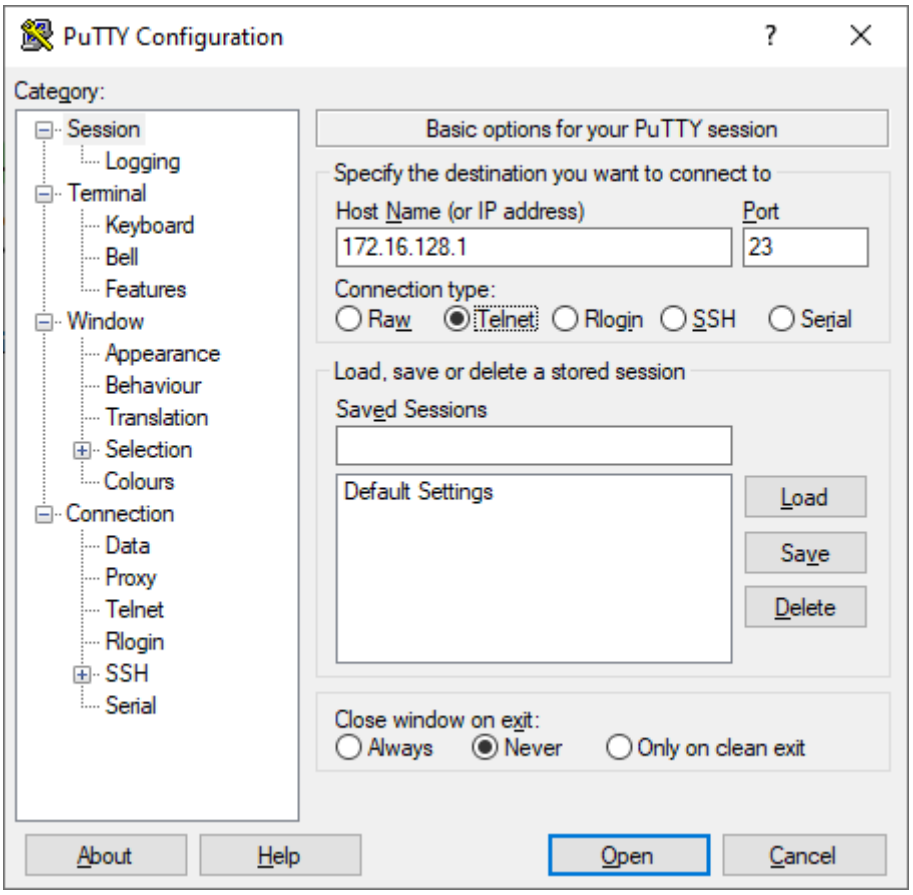


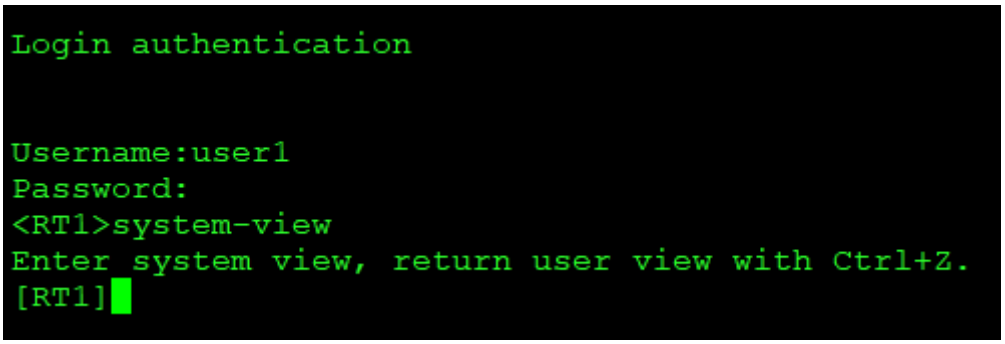

```
[RT1] display telnet server status
TELNET IPV4 server                          :Enable
TELNET IPV6 server                          :Enable
TELNET server port                          :23
```

```
[RT1] user-interface vty 0 4
[RT1-ui-vty0-4] authentication-mode aaa

[RT1] aaa
[RT1-aaa] local-user user1 password cipher user@123
[RT1-aaa] local-user user1 privilege level 15
[RT1-aaa] local-user user1 service-type telnet
```

Download PuTTY
https://putty.org/





3-қадам: Ubuntu Linux дистрибутивімен eNSP симуляторын байланыстыру

Download Linux images (OVF Files)
https://drive.google.com/drive/folders/1cUMzXEw4BFvp3dGhMJQiGjQnVS0b3MTB?usp=sharing
Download VMware Workstation Pro
https://drive.google.com/drive/folders/1xPeOKfdeOzGdEHJRhYktJThgL6-xjkHy?usp=sharing

Link YouTube Channel
https://www.youtube.com/playlist?list=PLqzHIBIxhTu2wlO2GhGKEV5edlE0z7SSb

...

login: student
Password: 123

student@ubuntu22-04~$ `ip address`

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 00:0c:29:a0:a0:d2 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.0.104/24 metric 100 brd 192.168.0.255 scope global dynamic
       valid_lft 7132sec preferred_lft 7132sec
    inet6 fe80::20c:29ff:fea0:a0d2/64 scope link
       valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 00:0c:29:a0:a0:dc brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet6 fe80::20c:29ff:fea0:a0dc/64 scope link
       valid_lft forever preferred_lft forever
```

student@ubuntu22-04~$ `sudo nano /etc/netplan/00-installer-config.yaml`

```
network:
  ethernets:
    ens33:
      dhcp4: true
    ens37:
      addresses: [172.16.128.5/24]
  version: 2
```

Ctrl+O ➜ Enter
Ctrl+X

student@ubuntu22-04~$ `sudo netplan apply`

student@ubuntu22-04~$ `ip address`

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 00:0c:29:a0:a0:d2 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.0.104/24 metric 100 brd 192.168.0.255 scope global dynamic
       valid_lft 5393sec preferred_lft 5393sec
    inet6 fe80::20c:29ff:fea0:a0d2/64 scope link
       valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 00:0c:29:a0:a0:dc brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    inet 172.16.128.5/24 brd 172.16.128.255 scope global ens37
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fea0:a0dc/64 scope link
       valid_lft forever preferred_lft forever
```

student@ubuntu22-04~$ `ping -c4 172.16.128.254`

```
64 bytes from 172.16.128.254: icmp_seq=1 ttl=128 time=0.453 ms
64 bytes from 172.16.128.254: icmp_seq=2 ttl=128 time=0.803 ms
64 bytes from 172.16.128.254: icmp_seq=3 ttl=128 time=0.466 ms
64 bytes from 172.16.128.254: icmp_seq=4 ttl=128 time=0.494 ms
```

student@ubuntu22-04~$ `ping -c4 172.16.128.1`

```
64 bytes from 172.16.128.1: icmp_seq=1 ttl=255 time=11.1 ms
64 bytes from 172.16.128.1: icmp_seq=2 ttl=255 time=7.69 ms
64 bytes from 172.16.128.1: icmp_seq=3 ttl=255 time=13.0 ms
64 bytes from 172.16.128.1: icmp_seq=4 ttl=255 time=9.76 ms
```

4-қадам: Python TELNET Library

student@ubuntu22-04~$ `telnet`

```
telnet> open 172.16.128.1
Trying 172.16.128.1...
Connected to 172.16.128.1.
Escape character is '^]'.


Login authentication


Username:user1
Password:
<RT1>system-view
Enter system view, return user view with Ctrl+Z.
[RT1]
```

student@ubuntu22-04~$ python3 --version

```
Python 3.10.6
```

student@ubuntu22-04~$ `ls -l`

```
-rw-rw-r-- 1 student student 318 Apr 22 09:58 script1.py
-rw-rw-r-- 1 student student 572 Apr 22 09:58 script2.py
-rw-rw-r-- 1 student student 967 Apr 22 09:59 script3.py
```

student@ubuntu22-04~$ `nano script1.py`

```
import telnetlib

host = '172.16.128.1'
user = 'user1'
password = 'user@123'

tn = telnetlib.Telnet(host)

tn.read_until(b'Username:')
tn.write(user.encode('ascii') + b'\n')
tn.read_until(b'Password:')
tn.write(password.encode('ascii') + b'\n')

print('Connection to ' + '172.16.128.1' + ' is Successful')

tn.close()
```

student@ubuntu22-04~$ `python3 script1.py`

```
Connection to 172.16.128.1 is Successful
```

student@ubuntu22-04~$ `nano script2.py`

```
import telnetlib
import time

host = '172.16.128.1'
user = 'user1'
password = 'user@123'
UserPrompt = '>'
ConfigPrompt = ']'

tn = telnetlib.Telnet(host)

tn.read_until(b'Username:')
tn.write(user.encode('ascii') + b'\n')
tn.read_until(b'Password:')
tn.write(password.encode('ascii') + b'\n')

print('Connection to ' + '172.16.128.1' + ' is Successful')

UserMode = tn.read_until(UserPrompt.encode('ascii'))
print(UserMode.decode('ascii'))

tn.write(b'system-view \n')
ConfigMode = tn.read_until(ConfigPrompt.encode('ascii'))
print(ConfigMode.decode('ascii'))

tn.close()
```

student@ubuntu22-04~$ `python3 script2.py`

```
Connection to 172.16.128.1 is Successful

<RT1>
system-view
Enter system view, return user view with Ctrl+Z.
[RT1]
student@ubuntu22-04:~$
```

student@ubuntu22-04~$ `nano script3.py`

```
import telnetlib
import time

host = '172.16.128.1'
user = 'user1'
password = 'user@123'
UserPrompt = '>'
ConfigPrompt = ']'

tn = telnetlib.Telnet(host)

tn.read_until(b'Username:')
tn.write(user.encode('ascii') + b'\n')
tn.read_until(b'Password:')
tn.write(password.encode('ascii') + b'\n')

print('Connection to ' + '172.16.128.1' + ' is Successful')

UserMode = tn.read_until(UserPrompt.encode('ascii'))
print(UserMode.decode('ascii'))
```

```
tn.write(b'system-view \n')
ConfigMode = tn.read_until(ConfigPrompt.encode('ascii'))
print(ConfigMode.decode('ascii'))

tn.write(b'interface LoopBack 0 \n')
ConfigMode = tn.read_until(ConfigPrompt.encode('ascii'))
print(ConfigMode.decode('ascii'))

tn.write(b'ip address 50.0.1.1 24 \n')
ConfigMode = tn.read_until(ConfigPrompt.encode('ascii'))
print(ConfigMode.decode('ascii'))

tn.write(b'display ip interface brief \n')
ConfigMode = tn.read_until(ConfigPrompt.encode('ascii'))
print(ConfigMode.decode('ascii'))

tn.close()
```

student@ubuntu22-04~$ `python3 script3.py`

```
Connection to 172.16.128.1 is Successful

<RT1>
system-view
Enter system view, return user view with Ctrl+Z.
[RT1]
interface LoopBack 0
[RT1-LoopBack0]
ip address 50.0.1.1 24
[RT1-LoopBack0]
display ip interface brief

Interface                         IP Address/Mask      Physical   Protocol
GigabitEthernet0/0/0              172.16.128.1/24      up         up
GigabitEthernet0/0/1              unassigned           down       down
GigabitEthernet0/0/2              unassigned           down       down
LoopBack0                         50.0.1.1/24          up         up(s)
NULL0                             unassigned           up         up(s)
[RT1-LoopBack0]
student@ubuntu22-04:~$
```

5-қадам: Python Netmiko (Paramiko) Library

...

6-қадам: Python NAPALM Library

...